

UPQC DC link, which enables extended voltage support during deep sags, temporary outages, and fault conditions. Another significant development is the use of wide bandgap semiconductor devices like silicon carbide (SiC) and gallium nitride (GaN), which allow UPQC converters to operate at higher switching frequencies with lower losses, resulting in smaller passive components, higher power density, and improved overall efficiency.

Advanced and intelligent control strategies, including artificial intelligence-based controllers, adaptive control, and model predictive control, are also emerging to provide faster dynamic response, better disturbance prediction, and improved compensation accuracy under varying grid conditions. Furthermore, future UPQC systems are moving toward modular and scalable designs, such as modular multilevel converter (MMC)-based UPQC, which offer higher reliability, easier maintenance, and flexible capacity expansion. With the addition of IoT-based monitoring and communication, UPQC can support real-time supervision, remote diagnostics, and integration with smart grid management systems. These advancements position UPQC as a key enabling technology for ensuring high power quality, system stability, and efficient energy utilization in future smart grids and modern power distribution networks.

CHAPTER- 3 CUSTOM POWER DEVICES

3.1 INTRODUCTION

Custom power devices are power electronics-controlled devices meant for distribution systems, improving power quality by dealing with problems like sags, swells, harmonics, etc., for industrial customers. There are several types of Custom Power devices. Among them, some are Active Power Filters (APF), Surge Arresters (SA), Battery Energy Storage Systems (BESS), Super Conducting Magnetic Energy Systems (SMES), Static Electronic Tap Changers (SETC), Solid State Fault Current Limiter (SSFCL), Solid State Transfer Switches (SSTS), Static VAR Compensator (SVC), Distribution Series Capacitors (DSC), Dynamic Voltage Restorer (DVR), Distribution Static Synchronous Compensators (DSTATCOM), Uninterruptible Power Supplies (UPS), Unified Power Quality Conditioner (UPQC). A power conditioning device installed in the medium voltage distribution network of an electrical utility serves the objective of providing protection to either an entire plant, an entire feeder, or a block of customers or loads. Such devices will have voltage ratings of input and output of 1 KV to 38 KV, and the rating of loads will be more than 500 kVA. This technique of installing a power conditioning device at such a level is termed as custom power. The classification of custom power devices can be done into two major categories:

1. Network configuring type,
2. Compensating type.

The **Network Configuring Type** of devices changes the configuration of the power system network for power quality enhancement. SSCL (Solid State Current Limiter), SSCB (Solid State Circuit Breaker), and SSTS (Solid State Transfer Switch) are the most representative in this category.

The **Compensating Type** devices are used for active filtering load balancing, power factor correction, and voltage regulation. The family of compensating devices includes DSTATCOM (Distribution Static compensator), DVR (Dynamic voltage restorer), and Unified power quality conditioner (UPQC). DSTATCOM is connected in shunt with the power system while DVR is a series connected device that injects a rapid series voltage to compensate for the supply voltage. UPQC is a combination of DSTATCOM and DVR. It injects a series voltage and shunt currents into the system.

Despite the many approaches used to reduce voltage sags and swells, employing a customized power apparatus is the most effective way to cater to different needs. Custom Power refers to the utilization of electronic power regulators in the distribution system to address various issues of power quality. It guarantees that consumers receive pre-specified levels of quality and reliability of power supply, which may incorporate either one or all of the following qualities: absence of interruptions, minimal phase imbalance, reduced harmonics in the load voltage, low flicker in the load voltage, tolerance to variations, magnitude and duration of over-voltage and under-voltages within certain limits, and poor power factor loads without any adverse effect on the terminal voltage.

3.2 CORE COMPONENTS AND

FUNCTIONALITY IN

SIMULATION

The simulation is structured to test how each filter handles specific power quality issues in a power system.

3.2.1 Active Filter (AF)

An Active Filter is a power electronic device used to improve power quality by removing harmonics, compensating reactive power, and balancing load currents or voltages. It uses switches like IGBTs/MOSFETs and a control system to inject compensating signals into the power system. Types of Active Filters:

- a) Shunt Active Filter
- b) Series Active Filter
- c) Hybrid Active Filter

Shunt Active Filter (SAF): The Shunt Active Filter (SAF), also known as the Parallel Active Filter (PAF), is the most used active filter in electrical power systems. It is connected in parallel with the nonlinear load at the Point of Common Coupling (PCC) and is mainly used for compensating current related power quality problems. In modern distribution systems, nonlinear loads such as rectifiers, variable frequency drives (VFDs), switched-mode power supplies, and inverters draw distorted and non-sinusoidal currents from the supply